



The Seam Between Day and Night

Student copy (project or hand out)

The following examines WASP-121 b, a planet so close to its star that one hemisphere faces permanent day and the other permanent night, and how astronomers have begun to read the differences along the boundary between them.

1 Eight hundred and fifty-eight light-years from Earth, a planet called WASP-121 b circles its star about once every thirty hours. It orbits so close that the star fills its sky. The heat has locked it in place, one face turned forever toward the fire, the other turned forever toward the dark, so the planet has no sunrise and no sunset, only a permanent day on one side and a permanent night on the other.

2 Between those two hemispheres runs a seam, and a team of astronomers at the Max Planck Institute for Astronomy has now read it edge by edge. On the day side the air reaches about 2,500 degrees Celsius, hot enough to turn metal and rock to vapor; on the night side it falls to around 700, still glowing but darker, a difference of more than 1,700 degrees across a single world. The boundary, the line astronomers call the terminator, is not one line here but two: a morning edge, where the dark turns toward light, and an evening edge, where the light fails into dark.

3 Reading a seam on a world that far away takes patience and a trick of geometry. As the planet crosses the face of its star, from Earth a small, regular eclipse, it turns by about thirty degrees, so the starlight that filters through its air passes through one longitude after another in sequence. That light carries a record of what it passed through. Split into its colors, it shows which gases lie along each edge and how hot they are. The two edges do not match.

4 The evening edge swallows more starlight than the morning edge. The reason is wind. Superheated gas pours off the day side on a steady eastward current and piles into the evening air; the air swells with the heat, widens, and blocks more light. The evening edge is also drier. There the temperature runs high enough to tear water molecules apart into the atoms that make them, hydrogen and oxygen, so less water survives to be seen. In its place the trace of carbon monoxide grows stronger toward the evening, a tougher molecule that holds together where water cannot.

5 What the measurements do not fully explain is how large the difference is. The models of these atmospheres predict a gentler imbalance than the telescope found, which means something in the evening air, some haze or cloud the equations do not yet carry, is still unaccounted for. The planet was found in 2015, the first beyond our own system shown to hold water and to wrap itself in a warm upper layer. A decade later it is read longitude by longitude: its locked face, its eastward wind, its swelling evening edge, its vanishing water and its surviving carbon monoxide, a single circling object that turns out to be made of differences, and not yet wholly known.

AP-style multiple choice:

1. The relationship between the third paragraph and the fourth paragraph is best described as one in which the writer first
 - A) raises a doubt about the measurements, then resolves it with new evidence
 - B) explains the method used to read the planet's atmosphere, then gives the physical cause of the difference that method revealed
 - C) describes the planet's discovery, then predicts what future telescopes will find
 - D) compares the planet to Jupiter, then explains why the comparison breaks down
2. In the final paragraph, the phrase 'not yet wholly known' primarily serves to
 - A) warn that the telescope's measurements are probably mistaken
 - B) imply that astronomers have lost interest in the planet
 - C) concede that the planet's atmosphere resists complete explanation even after close study
 - D) claim that the planet can never be studied any further

Discussion questions:

1. The writer accumulates physical detail about the planet but never says why the discovery matters or draws a lesson from it. What does the piece gain, and what might it give up, by withholding any judgment of significance?
2. Several times the writer states plainly what is not known, ending on a planet that is 'not yet wholly known.' How does admitting the limits of the account affect your trust in the writer?

Writing prompt (Q2 rhetorical analysis):

Read the passage carefully. Then write an essay that analyzes the rhetorical choices the writer makes to convey both the strangeness of WASP-121 b and the limits of what is currently known about it. Consider how the writer's accumulation of physical detail, contrast between short and long sentences, and admissions of uncertainty work together.